

Myra Cropper

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EDUCATION

Carnegie Mellon University: Pittsburgh, PA

Masters of Science in Electrical and Computer Engineering - Artificial Intelligence. *December 2026*

William & Mary: Williamsburg, VA

Bachelor of Science in Computer Science, Minor in Math. *May 2025*

Department Awards: Undergraduate Stephen Park Research Award, Elias Paparis Fellowship, Research Assistant Fellowship

SKILLS

Languages: Python; Java; C; C++

Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, ROS2, Hugging Face models

Cloud & MLOps: Azure, AWS, Docker, Kubeflow, Kubernetes, KServe, Model Context Protocol (MCP)

Databases & Storage: PostgreSQL, Firebase, ChromaDB

PROFESSIONAL EXPERIENCE

EverWatch – Booz Allen: Annapolis Junction, MD

AI/ML/Cloud Computing Intern (AI Sensemaking Platform, DroneVLM)

June 2025 – August 2025

- Engineered a 5-agent LangGraph pipeline with PostgreSQL, SentenceTransformers, and locally hosted DeepSeek-R1 LLM, ingesting 6,700+ OSINT records from social media and news into a ChromaDB vector store for semantic search
- Integrated Model Context Protocol (MCP) to coordinate multi-agent collaboration, enabling contextual narrative synthesis across domain-specific sub agents with NER, sentiment analysis, and geolocation tools for detecting sentiment spikes
- Optimized RAG retrieval using LangGraph by developing a prompt-driven proxy agent dynamically routing queries to domain-specific subagents, and optimized queries via LLM, improving response relevance and reducing query latency by 75%
- Programmed an autonomous DJI Tello drone with SmolVLM, PyTorch, and OpenCV to scan environments for physical cybersecurity threats, operating grid-based navigation and real-time image analysis for threat detection and logging

AI/ML/Cloud Computing (End-to-end MLOps Pipeline on AWS Kubernetes cluster)

June 2024 – August 2024

- Devised an end-to-end machine learning pipeline for MLOps Level 2 harnessing Kubeflow for malware and network intrusion detection with a 99% accuracy rate (From reading in data, EDA, to model serving with KServe)
- Proposed an object detection and classification pipeline with YOLO for a defense application with a 96% accuracy rate; Invented a web app interface performing real-time detection on user uploaded images
- Structured data storage adopting PostgreSQL for feature, model registry, and metadata storage

Hardwire: Pocomoke, MD.

Software Engineer Intern

May 2023 – August 2023

- Developed a Java URCap interface to bridge Universal Robots (UR3/UR5) with Cognex Vision Systems over TCP/IP (Telnet); automated a previously manual alignment process in production line
- Established scripts to monitor robots' acceleration through RTDE; decreased downtime of robots by 23%
- Communicated with team of three engineers to get PLC data; plotted data employing Blazor and C# on a web application

RESEARCH EXPERIENCE

Autonomous Vehicle Research, Vehicle Computing Lab: Williamsburg, VA.

First author: "Adaptive Multi-Camera Sensor Fusion for Autonomous Vehicles"

January 2025 – May 2025

- Designed and implemented a multi-sensor perception system fusing multi-camera Vision Transformer (ViT) features via Multi-Head Latent Attention (MLA) and models temporal dynamics with LSTM
- Achieved a 33% reduction in localization error and 25% gain in trajectory prediction accuracy on Canadian Adverse Driving Conditions (CADC) dataset
- Demonstrated improved performance over a Convolutional Neural Network (CNN) baseline achieving an accuracy of 75%

Honors Thesis: "Multi-Sensor Fusion and V2V Communication in Autonomous Vehicles: A Comprehensive Framework for Self-Driving" *January 2024 – December 2025*

- Fine-tuned EfficientNet model with Long Short-Term Memory (LSTM) with a 92% accuracy for detecting lane lines and predicting robot direction based on camera input; ROS2 to command differential-drive robot on docker container
- Implemented YOLO models with 94% accuracy for detection/object tracking on traffic signs, traffic lights, and other vehicles
- Utilized Data Distributed Services (DSS) via ROS2 for multi-robot communication for robot parking and intersection crossing
- Formulated camera data fusion for distance-based object detection; modeled robot environment Octomapping in RVIZ

PROJECTS

Canbus Spoofing

January 2025

- Developed a CAN bus spoofing tool using Arduino Uno and MCP2515 shield to inject modified speed and RPM values into a test vehicle
- Leveraged Opendbc files for signal decoding and Python cantool library to read default vehicle signals
- Configured Comma AI's Openpilot to run in CARLA, MetaDrive, and updated fingerprint to run on test vehicle

Comma AI's Calibration Challenge

January 2024

- Developed a Kalman filter-lane detection system for Comma AI's calibration challenging, achieving MSE of <25% on test data
- Implemented curved lane detection, pitch/yaw calculation from vanishing points, and speed estimation via optical flow, and parallelism enforcement for consistent lane tracking